FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2021

PAPER –1

TEST DATE: 06-06-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

  Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
  Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
  For each question, choose the option(s) corresponding to (all) the correct answer(s)
  Answer to each question will be evaluated according to the following marking scheme:
  Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
  Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
  Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
  Negative Marks **: –2** In all other cases.

  Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

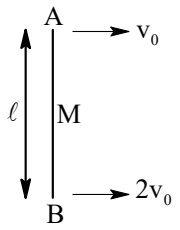
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A uniform rod of mass M and length ℓ is moving on a smooth horizontal plane, such that its one end is moving with a velocity v_0 and other end is moving with a velocity $2v_0$ in the same direction as shown. Then the kinetic energy of the rod is:



(A) $\frac{13}{24}Mv_0^2$

(B) $\frac{24}{31}Mv_0^2$

(C) $\frac{31}{24}Mv_0^2$

(D) $\frac{7}{6}Mv_0^2$

2. When a Plano wire is sounded together with a tuning fork which is vibrating at 440 Hz, three beats per second are observed. If the tension in the Plano wire is increased slightly, then the beat frequency increases. Find the initial frequency of the Plano wire?

(A) $434\,\text{Hz}$

(B) $437\,\text{Hz}$

(C) $443\,\text{Hz}$

(D) $446\,\text{Hz}$

3. Two particles of masses m and 2m has initial velocity $\vec{u}_1 = 2\hat{i} + 3\hat{j}\,\text{m/s}$ and $\vec{u}_2 = -4\hat{i} + 3\hat{j}\,\text{m/s}$ respectively. These particles have constant acceleration $\vec{a}_1 = 4\hat{i} + 3\hat{j}\left(\text{m/s}^2\right)$ and $\vec{a}_2 = -4\hat{i} - 2\hat{j}\left(\text{m/s}^2\right)$ respectively. Path of the centre of mass of this two particle system will be:

(A) Straight line
(B) Circular
(C) Parabolic
(D) Helical

4. If two uniform spherical drops of a liquid having same mass, radius, and falling freely under gravity with a terminal velocity of $\sqrt[3]{16}\,\text{m/s}$, coalesce to form a bigger drop, then the terminal velocity with which the bigger drop falls is (Neglect buoyancy force).

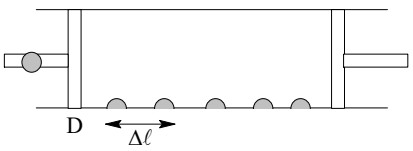
(A) $8\,\text{m/s}$

(B) $4\,\text{m/s}$

(C) $10\,\text{m/s}$

(D) $12\,\text{m/s}$

5. In a Kundt's tube distance between two consecutive heap is $\Delta\ell$ for air, while it is increased by 50% for a gas in the same tube with same resonate. If speed of sound in air is $\frac{1000}{3}\text{m/s}$ then speed of sound in gas at same temperature is.



(A) $1500\,\text{m/s}$

(B) $1000\,\text{m/s}$

(C) $500\,\text{m/s}$

(D) $250\,\text{m/s}$

6. A light ray is incident on a glass sphere of equation $x^2+y^2+z^2=a^2$. If the ray moves along the line $y\frac{a}{\sqrt{2}}$ from left to right in the x-y plane from outside the sphere. The angle of deviation suffered by the light ray as the ray comes out of the sphere is : (the refractive index of glass is $\mu=\sqrt{2}$).

(A) 60^0

(B) 48^0

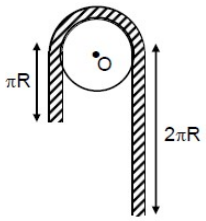
(C) 30^0

(D) depends upon 'a'

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. Consider a rope of mass 4m and length $4\pi R$ on a fixed rough pulley of radius R as shown in the figure. The rope is in equilibrium. Length of vertical hanging parts is shown in the figure.



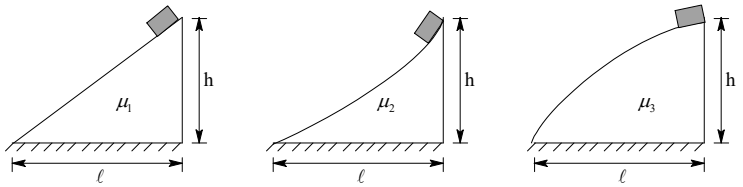
(A) Torque of tension force about O on pulley is 4mgR

(B) Torque of normal force between rope and pulley on pulley about O is zero.

(C) Torque of friction force between rope and pulley on pulley about O is mgR

(D) Torque of friction force between rope and pulley on pulley about O is zero.

8. Consider three fixed surfaces shown in the figure. Three blocks each of mass m are released from rest from top of the three surfaces. All blocks reach ground with same speed. Length of path travelled by the blocks is same for second and third surface. If coefficient of friction of three surfaces are μ_1,μ_2 and μ_3 respectively then:



(A) $\mu_1=\mu_2$

(B) $\mu_1 > \mu_2$

(C) $\mu_2 = \mu_3$

(D) $\mu_2 < \mu_3$

9. An uniform ring of mass m and radius R is gently kept at height h on an inclined plane of angle θ. If $\mu = \frac{\tan\theta}{4}$, then:

(A) Kinetic energy of the ring when it reaches the bottom is $\frac{2}{3}mgh$

(B) Time taken by the ring to reach the bottom is $\sqrt{\frac{8h}{3g\sin^2\theta}}$

(C) Acceleration of centre of mass of the ring is $\frac{3g\sin\theta}{4}$

(D) Angle by which ring rotates about its centre during the journey from top to bottom is $\frac{h}{3R\sin\theta}$ radians.

10. Two particles 1 and 2 are moving in concentric circular paths of radius R_1 and R_2 with constant speed. At a certain instant of time both are situated on the same radial line and have radial accelerations $\vec{a}_1, \vec{a}_2$ and tangential velocities $\vec{v}_1, \vec{v}_2$ respectively. Select correct alternative(s):

(A) Acceleration of 2 with respect to 1 is $\vec{a}_2 - \vec{a}_1$

(B) Acceleration of 2 with respect to 1 is $\vec{a}_2 - \vec{a}_1 + \frac{v_2^2}{R_2} - \frac{v_1^2}{R_1}$

(C) Angular velocity of 2 with respect to 1 is same as angular velocity of 1 with respect to 2

(D) Acceleration of 2 with respect to 1 is $\vec{a}_2 + \vec{a}_1 + \frac{v_2^2}{R_2} + \frac{v_1^2}{R_1}$

11. A convex lens forms an image of an object placed in its axis on a screen. The height of the image is $9\,cm$. Now the lens is displaced until an image is again obtained on the screen. The height of the image is $4\,cm$. The distance between the object and the screen is $90\,cm$. It follows that

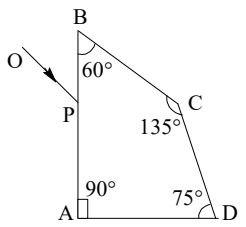
(A) The distance between the two positions of the lens is $30\,cm$

(B) The focal length of the lens is $24\,cm$

(C) The height of the object is $6\,cm$

(D) The distance of the object from the lens in its first position is $36\,cm$.

12. A ray OP of monochromatic light is incident at angle $60°$ on the face AB of prism ABCD near its vertex B as shown. The refractive index of the material of the prism is $\sqrt{3}$. Then, it follows that,



(A) The ray gets totally internally reflected at face CD

(B) The ray emerges through face AD

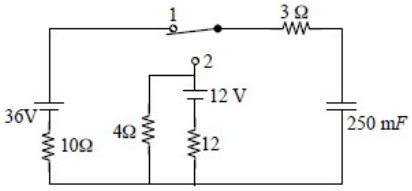
(C) The angle between the incident ray and the emergent ray is $120°$

(D) The angle of incidence of the ray with face AD is $30°$

SECTION – C
(Numerical Answer Type)

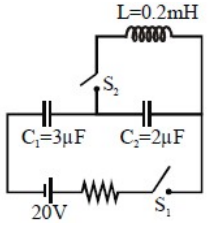
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. In the situation shown in figure, the switch is at position I for long time and its has shifted from position I, to position 2 at $t=0$. The current in the 3Ω resistor at $t=3$ ln 2 seconds is found to be $x\times\frac{11}{16}A$. Find the value of x.

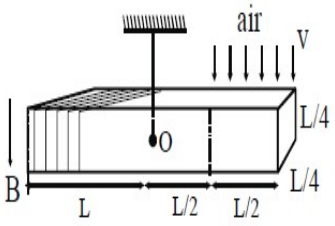


14. The binding energies per nucleon for a deuteron and an $\alpha-$particle are 1.1 MeV and $7.1\,\text{MeV}$ respectively. The energy released in the reaction ${}_1H^2+{}_1H^2\rightarrow{}_2He^4$ is Q. Then find the value of $\frac{Q}{3}$ in MeV.

15. The circuit shown in figure is in the steady state with switch S_1 closed and S_2 to open. At $t=0, S$ is opened and S_2 is closed. The first instant t, when the energy in the inductor becomes one-third of that in the capacitor C_2 is $\frac{\pi}{\alpha}$ millisecond. Find the value of α

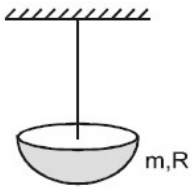


16. A non conducting non-magnetic rod having square cross section of side L/4 is suspended from a rigid support as shown in the figure. A light and small coil of 600 turns in wrapped tightly at the left end of the rod where uniform magnetic field B exists in vertically downward direction. Air of density ρ hits the half of the right part of the rod with velocity v as shown in the figure. What should be the current (in mA) in the coil so that rod remains horizontal?
[Assume air particles come to rest after colliding with the rod]
[Take: $\rho=1.05$ $\rho=1.05kg/m^3, v=10m/s, B=\text{IT, L=2m}$]



17. Ice at $0°\text{C}$ is added to $200\,\text{gm}$ of water initially at $70°\text{C}$ in a vaccum flask, having a constant heat capacity C. When $50\,\text{gm}$ of ice is added and has all melted the equilibrium temperature of the flask and its contents becomes $40°\text{C}$. Now when a further $80\,\text{gm}$ of ice is added and has all melted, the equilibrium temperature of the whole system becomes $10°\text{C}$. The latent heat of fusion of ice is L cal/gm, find $\frac{\text{L}}{10}$.

18. A uniform solid hemisphere of mass m and radius R is attached to the roof with a chord of torsional constant C and performing torsional SHM. Then the time period (in seconds) of SHM is (Take m = 15 kg, R = 2/ π m, C = 6 Nm/rad.)



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

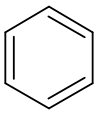
19. Which of the following ion is most stable in aq. solution?
(A) NH_4^+
(B) H_2F^+
(C) H_2O^+
(D) CH_5^+

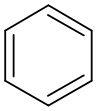
20. What are the possible oxidation state(s) of the Cr in the complex $K_2\left[Cr(NO)(NH_3)(CN)_4\right]$.
Given (magnetic moment) $\mu = 1.73\ BM$.
(A) +2
(B) +3
(C) +1
(D) All are possible

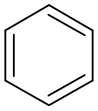
21. Which of the following statements concerning a weak acid-strong base titration is true?
(A) The pH after the equivalence point of the weak acid strong base titration is determined by using the K_b expression for the conjugate base
(B) A buffer solution of the weak acid and its conjugate base is formed before the equivalence point is reached
(C) The pH at the equivalence point of a weak monoprotic acid strong base titration is equal to the pH at the equivalence point of a strong acid-strong base titration
(D) The increase in pH in the region near the equivalence point of the weak acid strong base titration is greater than the pH change in the same region of a strong acid strong base titration

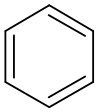
22. A hypothetical reaction
$A_2 + B_2 \longrightarrow 2AB$ Follows mechanism as given below:
$A_2 \underset{k_b}{\overset{k_c}{\rightleftharpoons}} A + A.....(fast)$
$A + B_2 \xrightarrow{k_1} AB + B.....(slow)$
$A + B \xrightarrow{k_2} AB.....(fast)$
The order of overall reaction is
(A) 2.5
(B) 1
(C) 3/2
(D) zero

23. If x is the number of chlorine atoms and y is the degree of unsaturation then which of these reactions will give a product with maximum value of $x+y$:

(A)  $+\underset{(\text{excess})}{CH_3COCl} \xrightarrow[AlCl_3]{\text{anhyd}}$

(B)  $+\underset{(\text{excess})}{Cl_2} \xrightarrow[\text{Cold, dark}]{\text{Anhyd. } AlCl_3}$

(C)  $+\underset{(\text{excess})}{CH_3Cl} \xrightarrow[AlCl_3]{\text{Anhyd.}}$

(D)  $+\underset{(\text{excess})}{Cl_2} \xrightarrow[500\,K]{h\nu}$

24. If the number of oxygen atom(s) present in hydrated (P) is:

$$(P) \xrightarrow{H_2SO_4} (Q) + BaSO_4\downarrow + H_2O$$

$$(Q) + H_2SO_4 + KI \longrightarrow K_2SO_4 + H_2O + (R)\downarrow$$

$$(R) \xrightarrow{\text{Starch}} \text{Blue colured}$$

(A) 10
(B) 18
(C) 16
(D) 15

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. $2CaSO_4(s) \rightarrow 2CaO(s) + 2SO_2(g) + O_2(g).\Delta H > 0$

Above equilibrium is established by taking sufficient amount of $CaSO_4(s)$ in a closed container at 1600 K. Then which of the following may be correct option (Assume that $CaSO_4$ is present in the container in each case)

(A) Moles of $CaO(s)$ will increase with the increase in temperature.

(B) If the volume of the container is doubled at equilibrium then partial pressure of $SO_2(g)$ will change at new equilibrium.

(C) If the volume of the container is halved partial pressure of $O_2(g)$ at new equilibrium will remain same

(D) If two moles of the He gas is added at constant pressure then the moles of $CaO(s)$ will increase

26. Which is / are **correct** statement?

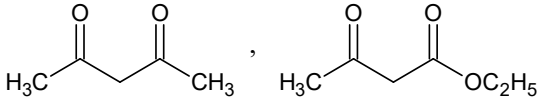
(A) $W_{\text{on the system, adiabatic}} > W_{\text{on the system, isothermally}}$ in an ideal gas compression from same initial state to same final volume

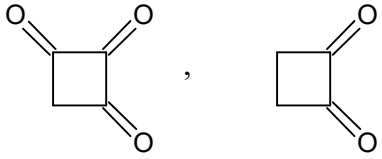
(B) The value of $\gamma \left(\gamma = \frac{C_p}{C_v} \right)$ remains constant for diatomic gas at all temperature

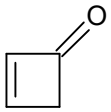
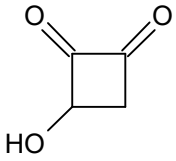
(C) Entropy increases when an ideal gas expanded isothermally.

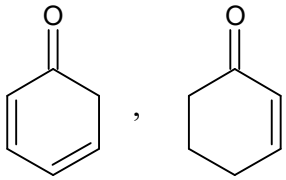
(D) $\Delta_r H$ & $\Delta_r S$ both are + ve for the decomposition of $MgCO_3(s)$.

27. In which of the following compounds, second has lesser enol content than first:

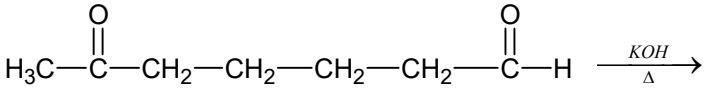
(A)  ,

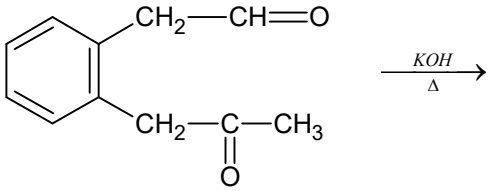
(B) 

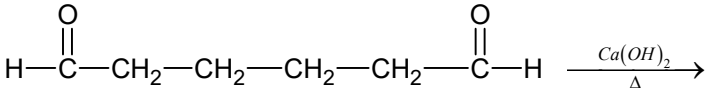
(C)  and 

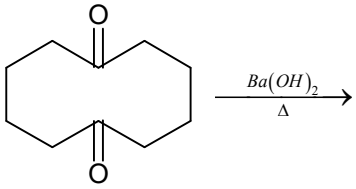
(D) 

28. In which of the following intramolecular aldol condensation reaction more than two products (stereoisomers are not considered) are possible?

(A) 

(B) 

(C) 

(D) 

29. Benzamide and aniline can be distinguished or reacted differently by:

(A) Ferric chloride test

(B) Carbyl amine test

(C) Dye azo test

(D) $NaOH/Br_2$

30. Select the correct statement:
(A) Lyophobic colloids are used to protect the lyophilic colloid
(B) Dehydration is used to coagulate the colloidal solution
(C) Rubber is obtained by coagulating the latex
(D) Sometimes heavy rainfall occurs when two oppositely charged clouds meets each other

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

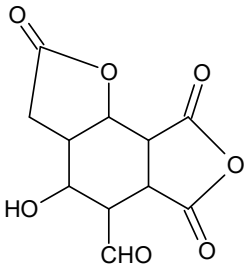
31. An equilibrium $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ is obtained by starting with1 mole of PCl_5 in a 1L vessel. A vapour density of equilibrium mixture was found to be 250.2. Now, the equilibrium mixture is transferred into a vessel of 2L capacity, its new degree of dissociation is.

32. A glow-worm of mass 5.0g emits red-light (650 nm) with power of 0.10 W entirely in the backward direction. What velocity (v) in m/sec, will it have accelerated to after 10 years, if released in free space (and assumed to live). Then v is?

33. In the unbalanced reaction:
$Al + KMnO_4 + H_2SO_4 \longrightarrow KHSO_4 + Al_2(SO_4)_3 + MnSO_4 + H_2O$
If the stoichiometric coefficients of $Al, H_2SO_4, MnSO_4 \text{and } H_2O$ are w, x, y and z respectively, the numerical value of $\left(\frac{x}{y} + \frac{z}{w}\right)$ will be

34. In acid rain in Delhi it was found that 1 litre rain water consist of 2.463 litre equimolar mixture of SO_3 and NO_2 gas dissolved at a partial pressure of 0.1 atm. Find out the pH of the solution if HNO_2 produced is 60% dissociated under given condition at $27°C$. (R = 0.0821 litre atm/mole K) (log 1.8 = 0.25)

35. If molecule shown below has molar mass→x then find the mass of CH_3MgBr (in grams) which can react (max.) with $\frac{x}{2}$ grams of compounds $[\text{Mg-24.3,Br-80}]$



36. $H_2C{=}C{=}O \xrightarrow[\text{(anddition) 1eq.}]{(1)\,Br_2} A \xrightarrow[2\text{eq.}]{CH_3MgBr} B$ suppose both reactions occur by 100% efficiency then the molar mass of B:

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Line L_1 : y – x + 2 = 0 strikes positive X-axis at a point A and gets reflected and becomes L_2 = 0. Line L_2 = O strikes again Y-axis and after reflection from Y-axis, it becomes L_3 = 0. Line L_3 = 0 meets the line L_4 : y + x – 4 = 0 at a point B. If the equation of the line passing through A and B be $y = mx + c$, then $c + m =$
(A) 6
(B) 3
(C) 2
(D) 4

38. Let a, b, c be positive integers such that $a+b+c = n$ then $\left(a^a b^b c^c\right)^{\frac{1}{n}} + \left(a^b b^c c^a\right)^{\frac{1}{n}} + \left(a^c b^a c^b\right)^{\frac{1}{n}}$ is less than or equal to:
(A) $\frac{ab + bc + ca}{n}$
(B) n
(C) abc
(D) None of these

39. From the point P (2, α), pair of tangents PA and PB are drawn to the extremities of the chord x + βy – 2 = 0 of the parabola y^2 + 2y – 8x + 33 = 0 If $\angle$ APB = $\pi/2$, then exhaustive set of values of α and β are
(A) $\alpha \in (0, \infty)$, $\beta \in R$
(B) $\alpha \in (-\infty, 0)$, $\beta \in (0, \infty)$
(C) $\alpha \in R$, $\beta \in R$
(D) $\alpha \in R$, $\beta \in (0, \infty)$

40. If $f : \mathbb{R} \to \mathbb{R}$ is a continuous function satisfying $f(0) = 1$ and $f(2x) - f(x) = x \ \forall x \in R$ and $\lim_{x \to \infty} \left(f(x) - f\left(\frac{x}{2^n}\right) \right) = P(x)$ then $P(x)$ is:
(A) A constant function
(B) A linear polynomial is x
(C) A quadratic polynomial is x
(D) A cubic polynomial is x

41. Let f(x) be a polynomial of degree n such that 1 + f(x) = $\frac{f(x-1) + f(x+1)}{2}$ $\forall$ x $\in$ R, then n satisfies
(A) n < 2
(B) n > 2
(C) n = 2
(D) None of these

42. Greatest integral value of 'n' for which the equation 2 cos x (sin x – cos x) = n has atleast one solution is

(A) -2
(B) 0
(C) 1
(D) 2

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. If a chord AB of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ subtends an angle $\frac{\pi}{2}$ at the origin then.

(A) Maximum value of $\left|\frac{1}{OA^2}-\frac{1}{OB^2}\right|=\frac{1}{a^2}+\frac{1}{b^2}$
(B) Maximum value of $\left|\frac{1}{OA^2}-\frac{1}{OB^2}\right|=\frac{1}{a^2}-\frac{1}{b^2}$
(C) Minimum value of $\left|\frac{1}{OA^2}+\frac{1}{OB^2}\right|=\frac{1}{a^2}-\frac{1}{b^2}$
(D) Area of ΔOAB is constant

44. If a, b, c $\in R^+$ and a + b + c = 8 then

(A) $\frac{a^2+b^2}{a+b}+\frac{b^2+c^2}{b+c}+\frac{c^2+a^2}{c+a}\geq 8$
(B) $\frac{a^2+b^2}{a+b}+\frac{b^2+c^2}{b+c}+\frac{c^2+a^2}{c+a}\geq 2$
(C) $a^2b^2 + b^2c^2 + c^2a^2 \geq 8abc$
(D) $a^2 + b^2 + c^2 \geq 18$

45. If $\alpha, \beta, \gamma, \delta, \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $\sin\alpha + \sin\beta + \sin\gamma + \sin\delta = 1$ and $\cos2\alpha + \cos2\beta + \cos2\gamma + \cos2\delta \geq 10/3$ then $\alpha, \beta, \gamma, \delta$ must lie in:

(A) $\left[0,\frac{\pi}{12}\right]$
(B) $\left[-\frac{\pi}{3},\frac{\pi}{3}\right]$
(C) $\left[0,\frac{\pi}{6}\right]$
(D) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

46. Let $f(x)=\begin{cases}[x]; & -2\le x<0\\ |x|; & 0\le x\le 2\end{cases}$; (where [.] denotes the greatest integer function) and $g(x)=\sec x \quad \forall\ x\in R-(2n+1)\frac{\pi}{2};\quad n\in I$. Then the points of discontinuity of fog is/are

(A) –1

(B) $\frac{5\pi}{6}$

(C) 2

(D) $-\pi$

47. Consider the equation $\frac{x^2}{\sin\sqrt{2}-\sin\sqrt{3}}+\frac{y^2}{\cos\sqrt{2}-\cos\sqrt{3}}=1$

($\sqrt{2}$ = 1.4142, $\sqrt{3}$ = 1.73205, π = 3.1415)

(A) The curve is an ellipse

(B) The curve is a hyperbola

(C) Major/Transverse axis is y–axis

(D) Major/Transverse axis is x–axis

48. A variable plane which remains at a constant distance P from the origin (0) cuts the coordinate axes in A, B, C.

(A) Locus of centroid of tetrahedron OABC is $x^2y^2 + y^2z^2 + z^2x^2 = \frac{16}{p^2}\ x^2y^2z^2$

(B) Locus of centroid of tetrahedron OABC is $x^2y^2 + y^2z^2 + z^2x^2 = \frac{4}{p^2}\ x^2y^2z^2$

(C) Parametric equation of the centroid of the tetrahedron is of the form $\left(\frac{P}{4}\sec\alpha\sec\beta,\frac{P}{4}\sec\alpha\,\mathrm{cosec}\beta,\frac{P}{4}\mathrm{cosec}\alpha\right)$ $\alpha, \beta \in (0, 2\pi) - \{\pi/2, \pi, 3\pi/2\}$

(D) None of these

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. $\int_0^3 (x^2+1)\,d([x])$ where [x] is the greatest integer less than or equal to x

50. Minimum value of $|z_1 + 1| + |z_2 + 1| + |z_1z_2 + 1|$ if $|z_1| = 1$, $|z_2| = 1$ is _____

51. If $\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{a}.\vec{b} & \vec{b}.\vec{b} & \vec{b}.\vec{c}\\ \vec{a}.\vec{c} & \vec{b}.\vec{c} & \vec{c}.\vec{c}\end{vmatrix} = 140$, then the volume of tetrahedron formed by $\vec{a},\vec{b}$ and $\vec{c}$ is equal to

52. If $|z| = 2$ and the maximum value of $|z-1|+|z-2|+|z-3|$ is M, then $\frac{M}{5}$ is equal to:

53. Let $\frac{1}{a_1-2i},\frac{1}{a_2-2i},\frac{1}{a_3-2i},\frac{1}{a_4-2i},\frac{1}{a_5-2i},\frac{1}{a_6-2i},\frac{1}{a_7-2i},\frac{1}{a_8-2i}$ are vertices of regular octagon. If the area of octagon is A. then the value of $\sqrt{2}\mathrm{A}$ is $=$
(Where $a_i \in \mathbb{R}$ for $i=1,2,....8$)

54. Normal at any point P on the ellipse $9x^2 + 16y^2 = 144$ meets the co-ordinate axes at A and B respectively, then maximum area of Δ OAB (O being origin) is_____